

1 / 20 | Ages 7-8 | 20 minutes

Paper bridge test

What we are practising

Change one feature and record an observable result.

What you need

Two equal sheets / Two books of equal height / Five large paper tokens / Pencil

Use small erasers as loads if no young children are nearby.

Before you begin

Put the books on a table with an 8 cm gap; keep this gap fixed.

An adult sets stable books on a low table; use light loads and never stand on a model.

Try it together

1. Lay one sheet flat across the gap and add tokens one at a time at its centre.
2. Record how many it supports before sagging into the gap, or record all five if it holds them.
3. Fold the second sheet into an accordion and repeat with the same gap and tokens.
4. Compare the results and draw the two bridge shapes.

Talk, draw or show your idea

What did you keep the same to make the comparison useful?

Space to explore

2 / 20 | Ages 7-8 | 15 minutes

Where does the water go?

What we are practising

Distinguish absorption from water sitting on a surface.

What you need

Tray / Cloth square / Paper towel square / Plastic lid / Teaspoon / Water

Compare just a cloth and a plastic lid.

Before you begin

Place dry samples on a tray away from electrical devices.

An adult supervises water use; wipe spills promptly and do not drink test water.

Try it together

1. Predict which sample will absorb water.
2. Put one teaspoon of water on each sample.
3. Wait about one minute, then look for spreading, soaking or a puddle.
4. Describe the evidence and dry the tray.

Talk, draw or show your idea

How can you tell soaked-in water from a puddle on top?

Space to explore

3 / 20 | Ages 7-8 | 15 minutes

Make a shadow grow

What we are practising

Relate shadow size to object position while keeping the light fixed.

What you need

Torch / Opaque book / Plain wall

Use a steady desk lamp with an adult positioning it safely.

Before you begin

An adult positions a cool torch facing a wall; keep the room bright enough to walk safely.

Never point a light into anyone's eyes; avoid hot lamps and keep walkways clear.

Try it together

1. Hold the book between the light and wall and observe its shadow.
2. Move the book toward the light without moving the torch.
3. Move the book toward the wall and compare the shadow sizes.
4. Explain which position made the bigger shadow.

Talk, draw or show your idea

What happens if the object blocks more of the spreading light?

Space to explore

4 / 20 | Ages 7-8 | 20 minutes

One place, many living things

What we are practising

Separate direct observations from questions about a habitat.

What you need

Paper / Pencil / Safe view of a garden or potted plant

Use a familiar plant indoors or a picture you already own.

Before you begin

Choose one place observable from a safe path or window.

An adult chooses a safe location; do not touch, taste, collect or feed unknown organisms.

Try it together

1. Spend two quiet minutes looking for plants and animals.
2. Draw three things you see, or draw fewer if only fewer are visible.
3. Label one observed feature, such as a leaf hole or a bird landing.
4. Write a question that your observation alone cannot answer.

Talk, draw or show your idea

Does not seeing an animal prove none live there?

Space to explore

5 / 20 | Ages 7-8 | 15 minutes

Flat or crumpled?

What we are practising

Test the effect of shape on falling through air.

What you need

Two identical sheets of scrap paper

Tear one sheet into two equal halves with adult help.

Before you begin

Clear a floor area and choose a drop height no higher than your shoulder.

Stay on the floor, never climb or drop objects from windows, and retrieve paper before walking.

Try it together

1. Leave one sheet flat and crumple the other tightly.
2. Predict which will land first.
3. Hold them at the same height and release together; repeat three times.
4. Compare results, including any release that was not simultaneous.

Talk, draw or show your idea

Did crumpling add any paper or change only its shape?

Space to explore

6 / 20 | Ages 7-8 | 10 minutes

Water outside a cold cup

What we are practising

Explain condensation without assuming a leak.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A sealed, leak-free cold cup develops droplets on its outside in a humid room.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A sealed, leak-free cold cup develops droplets on its outside in a humid room.
2. Ask: Did the droplets have to leak through the cup, or could they come from the air?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A sealed, leak-free cold cup develops droplets on its outside in a humid room. Did the droplets have to leak through the cup, or could they come from the air?

Space to explore

7 / 20 | Ages 7-8 | 10 minutes

Read the measuring scale

What we are practising

Read a scale with a non-unit interval.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a thermometer scale with marks 10,12,14,16,18,20 degrees Celsius; the indicator ends at 16.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw a thermometer scale with marks 10,12,14,16,18,20 degrees Celsius; the indicator ends at 16.
2. Ask: What temperature does it show, and how many degrees lie between neighbouring marks?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a thermometer scale with marks 10,12,14,16,18,20 degrees Celsius; the indicator ends at 16. What temperature does it show, and how many degrees lie between neighbouring marks?

Space to explore

8 / 20 | Ages 7-8 | 10 minutes

Which plant comparison is fair?

What we are practising

Identify a confounded comparison.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

To test light, plant A gets daylight and water; plant B gets darkness and no water.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. To test light, plant A gets daylight and water; plant B gets darkness and no water.
2. Ask: Would a difference in growth show that light alone caused it?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

To test light, plant A gets daylight and water; plant B gets darkness and no water. Would a difference in growth show that light alone caused it?

Space to explore

9 / 20 | Ages 7-8 | 10 minutes

Follow the food-chain arrow

What we are practising

Interpret a scientific arrow convention.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw grass, rabbit and fox; arrows point from food to the animal eating it.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw grass, rabbit and fox; arrows point from food to the animal eating it.
2. Ask: If the rabbit eats grass and the fox eats the rabbit, what is the arrow order?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw grass, rabbit and fox; arrows point from food to the animal eating it. If the rabbit eats grass and the fox eats the rabbit, what is the arrow order?

Space to explore

10 / 20 | Ages 7-8 | 10 minutes

Clear is not necessarily clean

What we are practising

Separate visible clarity from water safety.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Imagine muddy water passed through cloth; it looks clearer afterwards.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Imagine muddy water passed through cloth; it looks clearer afterwards.
2. Ask: Does clearer appearance prove the water is safe to drink?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Imagine muddy water passed through cloth; it looks clearer afterwards. Does clearer appearance prove the water is safe to drink?

Space to explore

11 / 20 | Ages 7-8 | 10 minutes

A hidden green insect

What we are practising

Explain camouflage through visual contrast.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a green insect on a green leaf and the same insect on plain white paper.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw a green insect on a green leaf and the same insect on plain white paper.
2. Ask: In which drawing would colour matching tend to make the insect harder to notice?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a green insect on a green leaf and the same insect on plain white paper. In which drawing would colour matching tend to make the insect harder to notice?

Space to explore

12 / 20 | Ages 7-8 | 10 minutes

Same clay, different boat

What we are practising

Connect floating with shape and displacement.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Equal clay lumps are shaped as a solid ball and a wide hollow boat; the boat floats while the ball sinks in a picture.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Equal clay lumps are shaped as a solid ball and a wide hollow boat; the boat floats while the ball sinks in a picture.
2. Ask: Does the floating boat prove the clay itself changed into a lighter material?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Equal clay lumps are shaped as a solid ball and a wide hollow boat; the boat floats while the ball sinks in a picture. Does the floating boat prove the clay itself changed into a lighter material?

Space to explore

13 / 20 | Ages 7-8 | 10 minutes

A pull through paper

What we are practising

Recognise a non-contact force from evidence.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

A magnet attracts an iron paper clip with a thin sheet of ordinary paper between them; use a drawing.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. A magnet attracts an iron paper clip with a thin sheet of ordinary paper between them; use a drawing.
2. Ask: Does a magnetic pull always require the magnet to touch the clip directly?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

A magnet attracts an iron paper clip with a thin sheet of ordinary paper between them; use a drawing. Does a magnetic pull always require the magnet to touch the clip directly?

Space to explore

14 / 20 | Ages 7-8 | 10 minutes

Find the circuit break

What we are practising

Explain an open circuit with a path model.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a simple battery-and-bulb loop with one wire disconnected; this is a diagram only.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw a simple battery-and-bulb loop with one wire disconnected; this is a diagram only.
2. Ask: In this simple circuit, why would the bulb not light despite a working battery and bulb?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a simple battery-and-bulb loop with one wire disconnected; this is a diagram only. In this simple circuit, why would the bulb not light despite a working battery and bulb?

Space to explore

15 / 20 | Ages 7-8 | 10 minutes

Seeds with travelling wings

What we are practising

Relate a structure to seed dispersal.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a winged seed and a plain pebble as objects to compare.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw a winged seed and a plain pebble as objects to compare.
2. Ask: How can a broad wing help some seeds travel on moving air?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a winged seed and a plain pebble as objects to compare. How can a broad wing help some seeds travel on moving air?

Space to explore

16 / 20 | Ages 7-8 | 10 minutes

Which side gets the shadow?

What we are practising

Predict shadow direction from the light source.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw a small lamp to the left of an opaque upright block, with a floor beneath.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw a small lamp to the left of an opaque upright block, with a floor beneath.
2. Ask: On which side of the block does the blocked-light region extend: towards or away from the lamp?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw a small lamp to the left of an opaque upright block, with a floor beneath. On which side of the block does the blocked-light region extend: towards or away from the lamp?

Space to explore

17 / 20 | Ages 7-8 | 10 minutes

Louder is not higher

What we are practising

Distinguish pitch from loudness.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Imagine the same musical note played softly and then loudly, without changing its pitch.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Imagine the same musical note played softly and then loudly, without changing its pitch.
2. Ask: Which property changes: loudness or pitch?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Imagine the same musical note played softly and then loudly, without changing its pitch. Which property changes: loudness or pitch?

Space to explore

18 / 20 | Ages 7-8 | 10 minutes

Settling is evidence

What we are practising

Limit a conclusion to visible settling evidence.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

In a picture, muddy water is left still; particles collect at the bottom while some remain suspended.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. In a picture, muddy water is left still; particles collect at the bottom while some remain suspended.
2. Ask: What can we safely conclude about the particles that reached the bottom?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

In a picture, muddy water is left still; particles collect at the bottom while some remain suspended. What can we safely conclude about the particles that reached the bottom?

Space to explore

19 / 20 | Ages 7-8 | 10 minutes

Where did the soil move?

What we are practising

Explain erosion and deposition as material movement.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw rainwater running down a bare slope, carrying soil grains into a lower tray.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw rainwater running down a bare slope, carrying soil grains into a lower tray.
2. Ask: Is the soil appearing from nowhere in the tray, or being moved from the slope?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw rainwater running down a bare slope, carrying soil grains into a lower tray. Is the soil appearing from nowhere in the tray, or being moved from the slope?

Space to explore

20 / 20 | Ages 7-8 | 10 minutes

Borrowed moonlight

What we are practising

Distinguish a light source from a reflector.

What you need

Paper / Pencil or crayons

An adult can describe the scene aloud and accept a spoken or pointed answer. No printer is needed.

Before you begin

Draw Sun, Moon and observer, without viewing the real Sun.

Explore the situation on paper with suitable drawing tools. Ask an adult before trying a real-world investigation.

Try it together

1. Draw Sun, Moon and observer, without viewing the real Sun.
2. Ask: Does the Moon make the visible light we call moonlight, or reflect sunlight?
3. Choose or draw an answer. Explain it using the clues; check the explanation together after trying.

Talk, draw or show your idea

Draw Sun, Moon and observer, without viewing the real Sun. Does the Moon make the visible light we call moonlight, or reflect sunlight?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. Paper bridge test

A little hint

Use the same number and position of loads for both bridges.

Answer / what to notice

Results vary; a folded sheet often bends less under the same load, but report what actually happened.

Folds change the paper's shape and resistance to bending without adding more paper.

Make it easier

Compare the two bridges with a single token.

Take it further

Repeat each shape three times and compare the results.

Another day, away from the screen

Look for folds or ridges in cardboard packaging.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Where does the water go?

A little hint

Look for a wet patch spreading through the material rather than a movable puddle.

Answer / what to notice

Cloth and paper towel usually soak up water; a solid plastic lid generally keeps it on the surface.

Many absorbent materials have tiny spaces that hold water; a smooth plastic surface does not soak it up in the same way.

Make it easier

Observe only one absorbent and one non-absorbent sample.

Take it further

Compare equal-sized samples of two cloths, repeating with fresh dry pieces.

Another day, away from the screen

Choose which tested material would help wipe up a small spill and explain why.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. Make a shadow grow

A little hint

Keep both the torch and wall in the same places.

Answer / what to notice

With a small fixed light source, the book usually makes a larger shadow closer to the light and a smaller one closer to the wall.

Light spreads out from the source, and the opaque object blocks part of that spreading light.

Make it easier

Compare only two marked book positions.

Take it further

Measure shadow width at three positions and record a table.

Another day, away from the screen

Make a short shadow story using your hands.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. One place, many living things

A little hint

Quiet or hidden animals may not appear during a short visit.

Answer / what to notice

Any accurate record is useful, including no visible animals; questions should be marked as questions.

A brief observation samples one place and time, so it cannot reveal everything living there.

Make it easier

Describe just one plant and its surroundings.

Take it further

Observe the same spot at a second time and compare without assuming a cause.

Another day, away from the screen

Keep a three-visit observation page and note weather or lighting.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. Flat or crumpled?

A little hint

A wide flat sheet pushes against more air as it falls.

Answer / what to notice

The crumpled sheet usually lands first; uneven release or drafts can affect a trial.

The sheets have the same amount of paper, but shape changes air resistance; mass alone does not explain this result.

Make it easier

Watch an adult release both sheets and describe what happened.

Take it further

Compare loose and tight crumpling while keeping paper size the same.

Another day, away from the screen

Sketch a paper shape that might fall slowly, then test it at the same safe height.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

6. Water outside a cold cup

A little hint

The air contains water vapour.

Answer / what to notice

They can come from water vapour in the air condensing on the cold surface.

Cooling nearby air can turn some vapour into liquid droplets; a leak is unnecessary.

Make it easier

Draw arrows from the surrounding air to the outside surface.

Take it further

Predict how a less humid room might change the amount of condensation.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

7. Read the measuring scale

A little hint

Read the marked value and the difference between neighbours.

Answer / what to notice

16°C; two degrees between marks.

The interval is found by subtraction, such as 14 minus 12.

Make it easier

Label the 16 mark clearly.

Take it further

Place the indicator halfway between 16 and 18 and estimate 17°C.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

8. Which plant comparison is fair?

A little hint

More than one condition changed.

Answer / what to notice

No. Water also differs; keep water and other conditions matched when comparing light.

Two changing factors make it hard to identify which caused the difference.

Make it easier

Circle the two differences in the plan.

Take it further

Redesign the comparison with similar plants and the same water, soil and observation period.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

9. Follow the food-chain arrow

A little hint

An arrow follows food energy into the eater.

Answer / what to notice

Grass to rabbit to fox.

Food transfers energy along the chain; the arrows here do not mean “chases”.

Make it easier

Draw just grass to rabbit first.

Take it further

Add another animal that eats grass and show a branching food web.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

10. Clear is not necessarily clean

A little hint

Some unwanted things are too small to see.

Answer / what to notice

No. Cloth may remove some visible particles without removing germs or dissolved substances.

Appearance cannot establish drinking safety; this is a picture investigation, with no tasting.

Make it easier

Name one invisible thing the cloth might not remove.

Take it further

Explain why water-treatment systems may need several different steps.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

11. A hidden green insect

A little hint

Compare contrast with the background.

Answer / what to notice

On the green leaf.

Similar colours can reduce visual contrast; movement and shape still matter.

Make it easier

Circle the insect on each background.

Take it further

Predict what changes if the leaf becomes brown.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

12. Same clay, different boat

A little hint

Shape and enclosed space changed.

Answer / what to notice

No. The hollow shape can displace enough water to support the same mass.

Floating depends on weight and displaced water, not only the material's name.

Make it easier

Compare the wide hollow space with the solid ball.

Take it further

Explain why filling the boat with water may make it sink.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

13. A pull through paper

A little hint

The paper separates the objects in the example.

Answer / what to notice

No; attraction can act across a small gap and through ordinary paper.

The observation is evidence of a force acting without direct contact.

Make it easier

Point to the separating paper.

Take it further

Predict why a much larger gap might weaken the effect.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

14. Find the circuit break

A little hint

Trace whether there is a complete conducting loop.

Answer / what to notice

The disconnected wire breaks the conducting loop.

Electric current needs a closed path in this basic circuit.

Make it easier

Trace from one battery terminal towards the other.

Take it further

Mark two different places where a break would stop the same circuit.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

15. Seeds with travelling wings

A little hint

A broad surface interacts with more moving air.

Answer / what to notice

It can slow their fall and help wind carry them away.

A wing changes how air acts on a seed; distance still depends on wind and seed shape.

Make it easier

Compare a broad wing with a compact shape.

Take it further

Suggest why different seed shapes may travel differently in the same breeze.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

16. Which side gets the shadow?

A little hint

Trace light from the lamp to the block.

Answer / what to notice

Away from the lamp, to the block's right in this drawing.

The opaque block prevents direct light from reaching the region behind it.

Make it easier

Draw a straight light ray stopped by the block.

Take it further

Move the lamp to the other side on paper and redraw the shadow.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

17. Louder is not higher

A little hint

Pitch describes high or low notes; loudness describes strong or soft sound.

Answer / what to notice

Loudness changes; pitch stays the same in this example.

These are different sound properties, so one can change without the other.

Make it easier

Compare the words high/low and loud/soft.

Take it further

Give an example of a high, quiet sound and a low, loud sound.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

18. Settling is evidence

A little hint

Report the change actually shown.

Answer / what to notice

They settled out of the water under these conditions.

The picture does not prove all particles settled or that the water is safe to drink.

Make it easier

Circle the material now at the bottom.

Take it further

Suggest taking pictures at equal intervals to compare settling over time.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

19. Where did the soil move?

A little hint

Follow the grains' route.

Answer / what to notice

It is being transported from the slope.

Moving water can erode and carry loose soil, then deposit it elsewhere.

Make it easier

Connect the slope and tray with an arrow.

Take it further

Predict how covering the slope with rooted plants could affect soil loss.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

20. Borrowed moonlight

A little hint

Compare a lamp with an object illuminated by a lamp.

Answer / what to notice

It reflects sunlight.

Sunlight reaches the Moon and some reflected light reaches our eyes.

Make it easier

Draw two arrows: Sun to Moon, then Moon to observer.

Take it further

Explain why a bright appearance alone does not prove an object makes its own light.

Another day, away from the screen

Draw the idea as a two-picture explanation and tell someone which part is observed and which part is explained.